# Support-Fins FS

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## 1 Overview

**Support-Fins FS** is a custom feature for your Onshape Part Studio toolbar. It adds breakaway supports to a part as real geometry, so the part can be printed in a strong tilted orientation with the slicer's own supports turned off. Because the supports are part of the model, they come out the same in any slicer, on any printer, in any filament.

Printing a part flat is often its weakest orientation; tilting it can make it several times stronger, but a tilted part normally needs slicer supports that scar the surface and are slow to pick off. This feature adds the supports that make the tilted orientation printable, in three kinds:

- **Overhang ribs** — thin upside-down-T walls under each overhang. The top of each rib follows the underside a small gap below it, and a comb of tiny **tines** reaches across that gap to hold the part.
- **Side bracing fins** — thin walls standing beside a flat side of a part that is tipped onto an edge or corner, gripping it with rows of tines so it cannot fall over.
- **Bed pad** — a thin oval brim under a part that touches the bed only along an edge or at a point, so it does not peel off.

The technique is Slant3D's designed-in "support fin". The feature is a port of the **support-fins** browser app (printfins.com, github.com/gittrahan/support-fins) to native Onshape geometry.

***Note:*** *The feature needs solid parts: native Onshape parts or imported STEP, Parasolid and similar CAD files. STL and other mesh files import into Onshape as mesh bodies, which it cannot process. For an STL, use printfins.com instead.*

## 2 Step-by-Step Workflow

Complete these steps in order:

| | |
|---|---|
| **1** | Add the feature to your toolbar once: **Custom features** (toolbar) → open the **Fin Supports** document → select **Support-Fins FS**. |
| **2** | In your Part Studio, orient the part the way it will sit on the printer, for example with a **Transform** feature. By default the **Top** plane is the build plate. |
| **3** | Open **Custom features** → **Support-Fins FS** and select the part under **Parts to support**. |
| **4** | Under **Print settings**, set **Layer height** to your slicer's layer height, **Nozzle** to your printer's nozzle size, and, for PETG, widen the **Support gap**. |

| | |
|---|---|
| **5** | Check the preview. Turn on **Show detected overhangs** to see which faces were found. Adjust **Overhang ribs**, and turn on **Side bracing fins** if the part rests on an edge or corner. |
| **6** | Click **OK**. The supports appear as orange parts named *<part> support 1, 2 …*, and a message reports what was built. |
| **7** | Select the part **and** all of its supports, right-click → **Export**, choose STL, and export them as one file. |
| **8** | Slice with supports **off**. After printing, bend each support sideways to snap it off. |

# 3 Feature Dialog Options

## 3.1 Selections

These sit at the top of the dialog, above the grouped settings.

| Option | Description |
|---|---|
| **Parts to support** | One or more solid parts, already positioned the way they will print. Each part gets its own supports. The parts themselves are never modified. |
| **Build plate** | A plane, flat face or mate connector whose direction is the printer bed. Leave empty to use the Top plane. Up is taken to be the side the part is on. |
| **Flip build direction** | Reverses which way is up, if the supports come out on the wrong side. |
| **Seat part on plate** | On by default. Treats the part's lowest point as the bed surface, so only the plate's direction matters, not its position. Turn off to use the plate exactly where it is. |

## 3.2 Overhang Ribs

This group controls the ribs placed under overhanging faces.

| Option | Description |
|---|---|
| **Auto-detect overhangs** | Finds downward faces too flat to print unsupported and puts ribs under them. |
| **Overhang angle (from plate)** | Downward faces flatter than this, measured from the bed, get ribs. Default 45°, the usual limit for printing without support. Raise it for more supports. |
| **Exclude faces** | Detected faces that should not get ribs, such as a cosmetic surface or one that bridges well on its own. |
| **Add overhang faces** | Faces to support even if detection skipped them. Any face that points at all downward can be added. Added faces also accept shorter ribs (4 mm instead of 7 mm). |
| **Max unsupported span** | The widest stretch of overhang left between two ribs. A wide overhang gets a row of ribs no farther apart than this. Default 12 mm (0.5 in). |
| **Gripping tines** | On by default. Tiny one-layer bridges from the rib top to the part. Off gives plain breakaway ribs that the part simply rests on. |

| Option | Description |
|---|---|
| **Tine spacing** | Distance between tines near the ends of each rib; the middle is spaced twice as far, up to 5 mm. Smaller grips harder but leaves more marks. Default 2 mm. |
| **Max walls per part** | A safety cap on how many ribs one part can get. Default 60. |

## 3.3 Side Bracing Fins

Collapsed and off by default. See Section 4.3 for when to use them.

| Option | Description |
|---|---|
| **Side fins** | **Off**, **Auto** (the feature picks the faces) or **Selected faces** (you pick them). |
| **Max fin sites** | Auto only. How many sides of the part may get a fin. The default of 2 braces opposite sides, which is usually enough. |
| **Faces to brace** | Selected faces only. The flat sides to put a fin against. Pick faces on opposite sides so the part cannot tip either way. |
| **Max face lean (from vertical)** | The steepest face lean, measured from vertical, that can take a fin. Default 45°. The fin leans with its face, so past 45° the fin itself becomes an overhang. |

## 3.4 Bed Pad

| Option | Description |
|---|---|
| **Bed pad** | **Auto** adds a pad only when the part has less than 60 $mm^2$ of flat contact with the bed, meaning it rests on an edge or a point. **Always** and **Never** override that. |

## 3.5 Print Settings

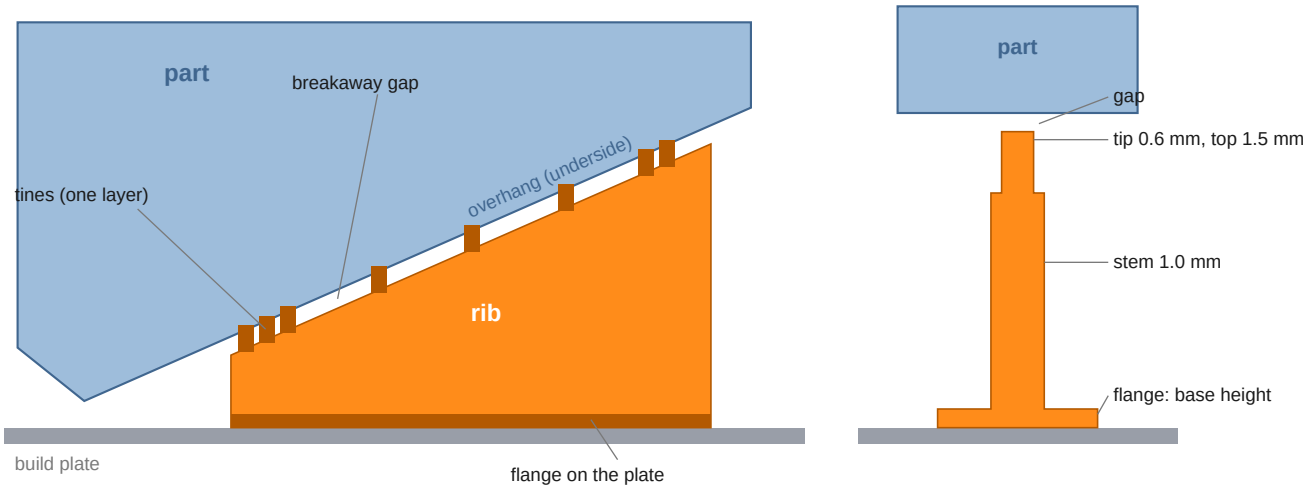
| Option | Description |
|---|---|
| **Layer height (tine height)** | Set this to your slicer's layer height. Each tine is exactly one layer tall and lined up with the layers, so it prints as a single strand that snaps clean. |
| **Nozzle** | Your printer's nozzle diameter (default 0.4 mm). Tines and contact tips are one bead wide, so they scale with it both ways; the ribs and side fins only get thicker for a bigger nozzle. |
| **Support gap** | Air gap between the supports and the part (default 0.2 mm, the website's). Bigger comes off easier and leaves a rougher underside. PETG sticks to supports much harder than PLA: try 0.3 mm. |
| **Show detected overhangs** | Highlights the overhang faces that were found, in red. |

# 4 How the Supports Are Built

## 4.1 Overhang ribs

Each downward face flatter than the overhang angle is found, faces that touch are grouped into one overhang, and overhangs smaller than 12 $mm^2$ are ignored. For each overhang:

- **Direction.** On a sloped underside the ribs run down the slope; on a nearly flat one they run along its longer side. A large curved underside, such as a tube lying down, gets a single rib under its lowest line.
- **Contour.** Along each rib the underside is sampled every 1 mm by firing a ray straight up from the plate. A rib is only placed where the first thing the ray hits is the overhang itself, so the path to the plate is clear.
- **Clearance.** A copy of the part, grown outward by the support gap, is subtracted from every rib. Whatever is left cannot touch the part anywhere, including the flanks.
- **Size limits.** Ribs shorter than 7 mm are skipped. Under 1.5 mm tall a rib becomes a squat wall on a thin, wide brim; under 0.6 mm the first layers print without help and nothing is added.




Side view: the rib top follows the underside

Cross-section: an upside-down T

*Figure 1. An overhang rib: side view (left) and cross-section (right).*

## 4.2 Tines

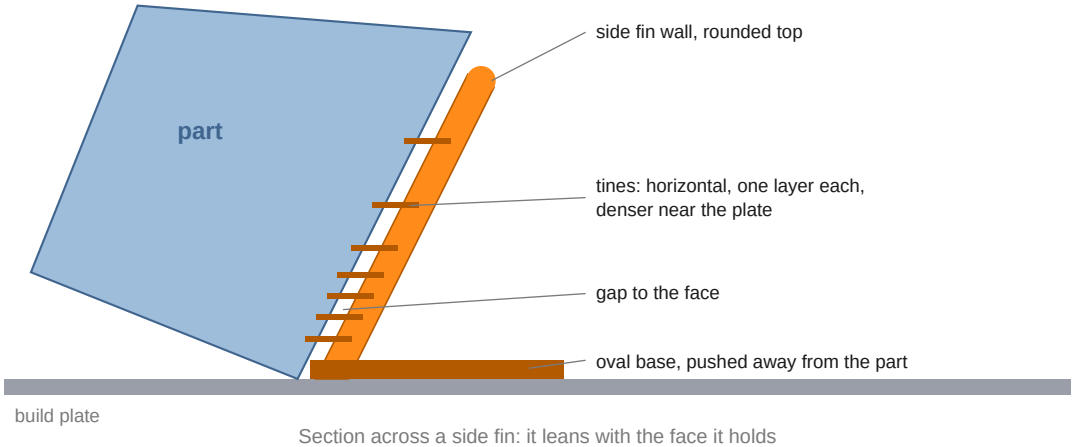
A tine is a small block one layer tall and 0.5 mm wide. It sits on the rib's top, sinks 0.3 mm back into the rib, and reaches across the gap to the part's surface, where it stops. Because it lies flat in a single layer, the nozzle prints it as one continuous strand: rib, tine, part and back, with no retraction. That strand is strong enough to hold the part, and it bends and snaps cleanly when the rib is removed.

- Each tine is snapped to the layer grid, so it never straddles two layers.
- A tine is only added where the part's solid lies within 0.5 mm of it. On a nearly flat underside a sideways reach finds nothing to grip, so flat overhangs get plain ribs.
- A tine ends at the part's surface at mid-layer, where the slicer cuts, as on the site. It no longer runs into the part, so it comes off cleaner.
- Tines are packed tightly within 8 mm of each end of a rib, where the marks hide on edges and corners, and spaced out across the middle. Every rib gets at least three.

## 4.3 Side bracing fins

A part tipped onto an edge or corner has a tiny footprint and can be knocked over by the nozzle or peel off partway through the print. Ribs only push up from underneath, so they do little to stop it tipping. A side fin holds the part from beside it instead:

- A 1.2 mm wall stands parallel to a flat side of the part, one support gap away, on an oval base 9 mm deep. If the side leans, the fin leans with it.
- Rows of horizontal tines cross the gap to the side and fuse to it: eight rows packed into the first 6 mm above the plate, then spreading out. The part is least stable near the bed.
- The fin is at most 25 mm long and only as tall as the face reaches across its whole length, so the entire fin has face behind it. On a tilted square face, which sits diamond-shaped, it goes in the middle where the face is tallest.
- A fin is abandoned, not shortened, if it or its base would touch another part of the model, or if fewer than three tines can grip.




Section across a side fin: it leans with the face it holds

*Figure 2. A side fin standing beside a leaning face.*

**When to use them.** Turn side fins on when the part balances on an edge or corner, and pair them with the bed pad. **Auto** ranks the flat faces by how much grippable wall they offer and picks up to **Max fin sites** faces pointing at least 60° apart and 12 mm apart. **Selected faces** tries every face you pick and reports any it could not use, with the reason. A face 55 mm or wider gets a row of fins about 55 mm apart instead of one.

> ***Note:*** *A face can take a fin if it is flat, at least 4 × 4 mm, leans no more than the* ***Max face lean*** *from vertical, reaches down near the bed, and has open space beside it.*

## 4.4 Bed pad

The pad is an oval fitted around where the part touches the bed: long and thin under an edge, round under a point, reaching 4 mm past the contact. Away from the part it is full height for bed grip; under the part its top follows the underside and overlaps it by the pad grab, a light tack that holds the part down but peels off afterward. The pad is the same height as the rib flanges, so where the two meet they merge flush, with no step.

## 5 Output

Supports are added as new parts. The original part is never modified.

| Item | Details |
|---|---|
| **Support parts** | Named *<part> support 1, 2 …* and colored orange. Ribs, tines, fins and pad that touch are merged into one part; separate supports stay separate. |
| **Feature message** | Reports the number of overhangs found, ribs, tines, side fins and pads, plus any overhangs left unsupported and why. |
| **Export** | Export the part together with all of its supports as **one STL**. The supports overlap the part slightly at the tines and pad; the slicer merges them into one solid. |

***Note:*** *If you export a 3MF or separate files, the slicer may load the supports as separate objects. Merge them into one object, or export as a single STL, so the tines fuse to the part.*

## 6 Printing and Removal

- Slice with the slicer's supports **off**. The model already carries its supports.
- Match the slicer's layer height, *and* its first-layer height, to the **Layer height** setting so every tine lands on exactly one layer.
- To remove a rib or fin, bend it sideways, in the plane of the layers. The tines fatigue and snap, leaving small, faint marks. Pulling straight off tears them out instead.
- Peel the bed pad off from its outer edge toward the part.
- For PETG, set **Support gap** to 0.3 mm. PETG bonds to supports far harder than PLA does.

## 7 Troubleshooting

The feature reports what it built, and anything it could not build, in a message on the feature. The table lists each message with its cause and what to do.

| Message | Cause | Remedy |
|---|---|---|
| **Select at least one part to support.** | Parts to support is empty. | Select a solid part. |
| **No supports generated.** | Nothing needed or could take a support at these settings. | Check Show detected overhangs. Raise the overhang angle, add faces manually, or turn on side fins. |
| **N region(s) got no wall** | An overhang was too small, too low, or over another part of the model rather than the plate. | Expected for features that bridge. Use Add overhang faces to force a short rib, or reorient the part. |

| Message | Cause | Remedy |
|---|---|---|
| **Part extends below the selected plate** | Seat part on plate is off and the part dips below the plate. | Turn Seat part on plate on, or move the plate. |
| **Could not offset the part for clearance** | The part could not be grown by the gap, so only a vertical gap was cut. | Check that the rib sides stand clear of the part. |
| **Clearance cut failed; overhang ribs were removed.** | The clearance cut failed on this geometry. | Simplify small features near the overhang, or try a slightly different gap. |
| **Supports could not be merged** | The support pieces could not be combined; tines remain separate parts. | Export them all together; the slicer will merge them. |
| **No side fin fit** | No face met the side fin requirements (Section 4.3). | Select faces manually, or raise Max face lean. |
| **Side fin, face N: leans more than the Max face lean** | The face is too far from vertical. | Raise Max face lean, or brace a different face. |
| **Side fin, face N: the fin or its foot would touch another part of the model** | There is no open space beside the face. | Pick a face on the outside of the part. |
| **Side fin, face N: fewer than 3 tines could grip the face** | The face is too small or oddly shaped for a grip. | Pick a larger face. |
| **Side fin, face N: face is too close to the bed / too short / too narrow** | Not enough face to stand a fin against. | Pick a taller, wider face. |

***Note:*** *Faces in the side fin messages are numbered in the order Onshape returns your selection.*

## 8 Tips and Best Practices

- Orient first, then support. The feature never chooses the orientation. Only you know which way the part will be loaded, and so which orientation makes it strong.
- Put overhangs on edges and corners where you can. That is where the tines are packed, and where their marks are hardest to see.
- A part whose underside sits at exactly 45° counts as self-supporting. Raise the overhang angle a degree or two to add ribs to it anyway.
- Long parts want support on both sides. With side fins, brace opposite faces so the part cannot twist.
- If the ribs leave too many marks, increase **Tine spacing**; if the part shifts during printing, decrease it.
- Keep the Layer height setting in step with the slicer. A tine that spans two layers bonds harder and marks worse.

## 9 Credits and License

Fin geometry, dimensions and placement rules are ported from **support-fins** by Matthew Trahan (github.com/gittrahan/support-fins, MIT License), which automates Slant3D's designed-in support fin technique.

This FeatureScript is released under the **MIT License**.

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